

IT Security 2

Unit 14 - Locating Valuable Information using Logs and Reports

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Announcements

Locating Valuable Information using Logs and Reports

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

LOCATING VALUABLE INFORMATION USING LOGS AND REPORTS



KEEP YOUR EYES ON THE ROAD

- View threat and traffic information:
 - In the Dashboard
 - In the ACC
 - In the logs
 - In App Scope reports
 - In predefined reports
 - In custom reports
- Forward threat and traffic information to external services

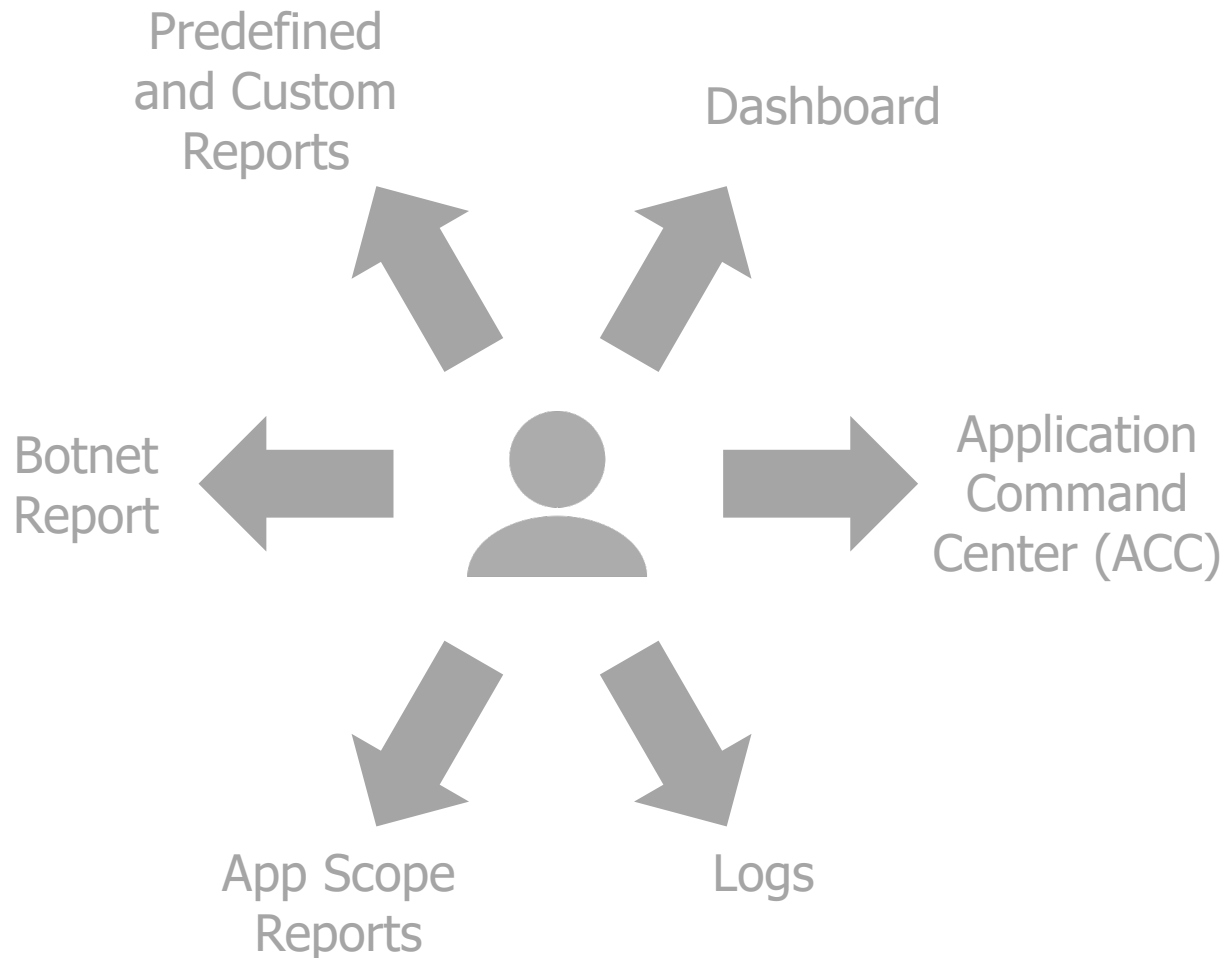
Learning Objectives

After you complete this module, you should be able to:

- Monitor threat and traffic information using the Dashboard and the ACC
- Monitor threat and traffic information using the logs
- Monitor threat and traffic information using App Scope reports
- Monitor threat and traffic information using predefined and custom reports
- Configure firewall log forwarding to external services



View Threat and Traffic Information



- Information is viewable in graphical and textual formats.
- Most information is viewable in the web interface.
- Some information is exportable and viewable outside the firewall.

View threat and traffic information:



In the Dashboard

In the ACC

In the logs

In App Scope reports

In predefined reports

In custom reports

Forward threat and traffic information to external services

The Dashboard

- The **Dashboard** consists of a customizable set of widgets.
- A widget is a tool that displays firewall information in a pane.

The screenshot shows the Palo Alto VM Dashboard interface. The top navigation bar includes tabs for DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. Below the navigation bar, there's a 'Layout' section with a '3 Columns' dropdown and a 'Widgets' dropdown menu. The main content area is divided into several widget panes: 'General Information' (labeled A), 'Interfaces' (labeled A), 'System Resources' (labeled A), 'ACC Risk Factor' (labeled A), 'Locks', and 'Config Logs'. A 'Logged In Admins' table is also visible. Annotations are placed on the interface: 'A' marks various example widgets; 'B' points to the 'Widgets' dropdown menu; 'C' points to the 'X' icon in the 'Logged In Admins' table header; and 'D' points to the 'Config Logs' widget. A text box on the left explains the annotations.

A: Example widgets.
B: Add a widget.
C: Remove a widget.
D: Drag and drop to rearrange widgets.

admin | Logout | Last Login Time: 07/23/2020 16:32:23 | Session Expire Time: 08/26/2020 15:44:42

Widgets for Viewing Threat Information

Three **Dashboard** widgets are used to display threat information.

The screenshot shows the PA-VM Dashboard with three widgets displayed in a 3-column layout. The top navigation bar includes tabs for DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. The top right corner shows a 'Commit' button and a search icon. The top left corner shows the PA-VM logo. The top bar also includes a 'Layout' dropdown set to '3 Columns', a 'Widgets' dropdown, and a 'Last updated' timestamp of '17:34:42'. The top right corner shows a '5 mins' refresh interval and a search icon.

Data Logs

File Name	Name	Time
fontawesome-webfont.woff2	WOFF File Format	07/27 21:33:00
page.php	Hypertext Preprocessor PHP File	07/27 21:31:59
return_js.php	Hypertext Preprocessor PHP File	07/27 21:31:59

URL Filtering Logs

URL	Category	Time
hacker9.com/favicon.ico	hacking	07/27 21:33:47
hacker9.com/login/css/latofonts.css	hacking	07/27 21:33:46
hacker9.com/	hacking	07/27 21:33:46
hidester.com/favicon.ico	proxy-avoidance-and-anonymizers	07/27 21:32:11
hidester.com/login/css/latofonts.css	proxy-avoidance-and-anonymizers	07/27 21:32:11
hidester.com/	proxy-avoidance-and-anonymizers	07/27 21:32:11
hidester.com/favicon.ico	proxy-avoidance-and-anonymizers	07/27 21:31:59
hidester.com/login/css/latofonts.css	proxy-avoidance-and-anonymizers	07/27 21:31:59
hidester.com/	proxy-avoidance-and-anonymizers	07/27 21:31:59
www.theguardian.com	proxy-avoidance-and-anonymizers	07/27 21:31:59

Threat Logs

Name	Severity	Time
Suspicious Domain	medium	07/27 21:34:39
Suspicious Domain	medium	07/27 21:34:38
Suspicious Domain	medium	07/27 21:34:35
Suspicious Domain	medium	07/27 21:34:33
Suspicious Domain	medium	07/27 21:34:33
Suspicious Domain	medium	07/27 21:34:31
Suspicious Domain	medium	07/27 21:34:28
Suspicious Domain	medium	07/27 21:34:27
Suspicious Domain	medium	07/27 21:34:22
Suspicious Domain	medium	07/27 21:34:22

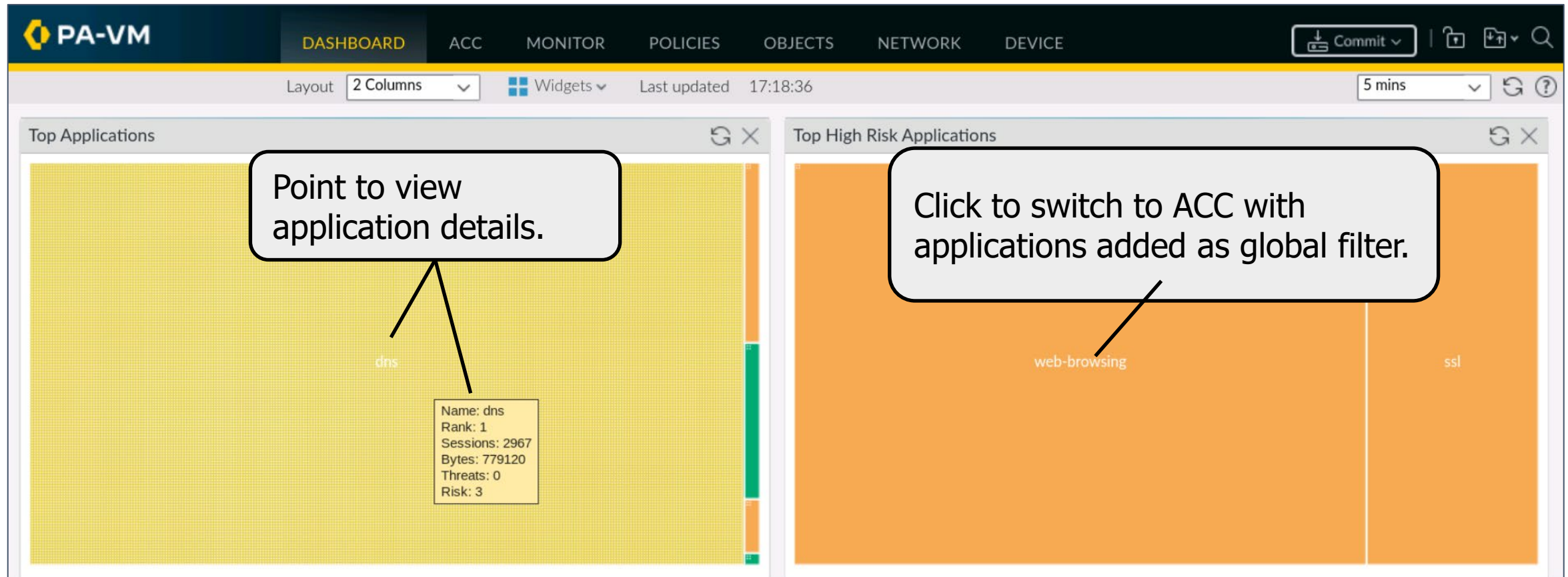
Last 10 Data Filtering log entries in the last hour

Last 10 URL Filtering log entries in the last hour

Last 10 Threat log entries in the last hour

Widgets for Viewing Application Information

Two **Dashboard** widgets are used to display application information.



View threat and traffic information:

In the Dashboard

In the ACC

In the logs

In App Scope reports

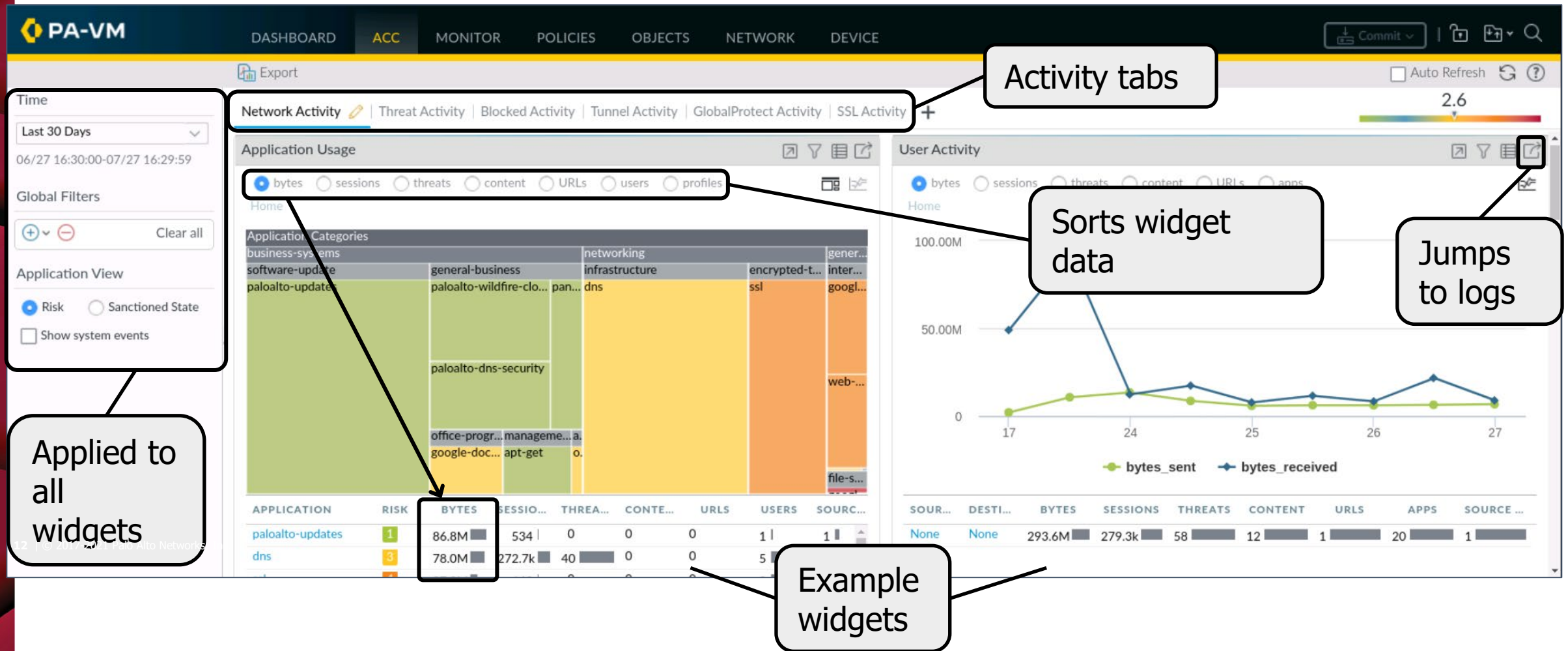
In predefined reports

In custom reports

Forward threat and traffic information to external services



Application Command Center (ACC)



For a Tech Doc about this topic, log into Live and search for "Use the Application Command Center"

Widgets on the ACC Network Activity and Threat Activity Tabs

These tabs are typically the most frequently used to view and analyze traffic and threat information.

Primarily used to view *application* and *traffic* information

Available Widgets

- **Application Usage**
- **User Activity**
- **Source IP Activity**
- **Destination IP Activity**
- **Source Regions**
- **Destination Regions**
- **GlobalProtect Host Information**
- **Rule Usage**

Network Activity | Threat Activity

Primarily used to view *threat* information

Available Widgets

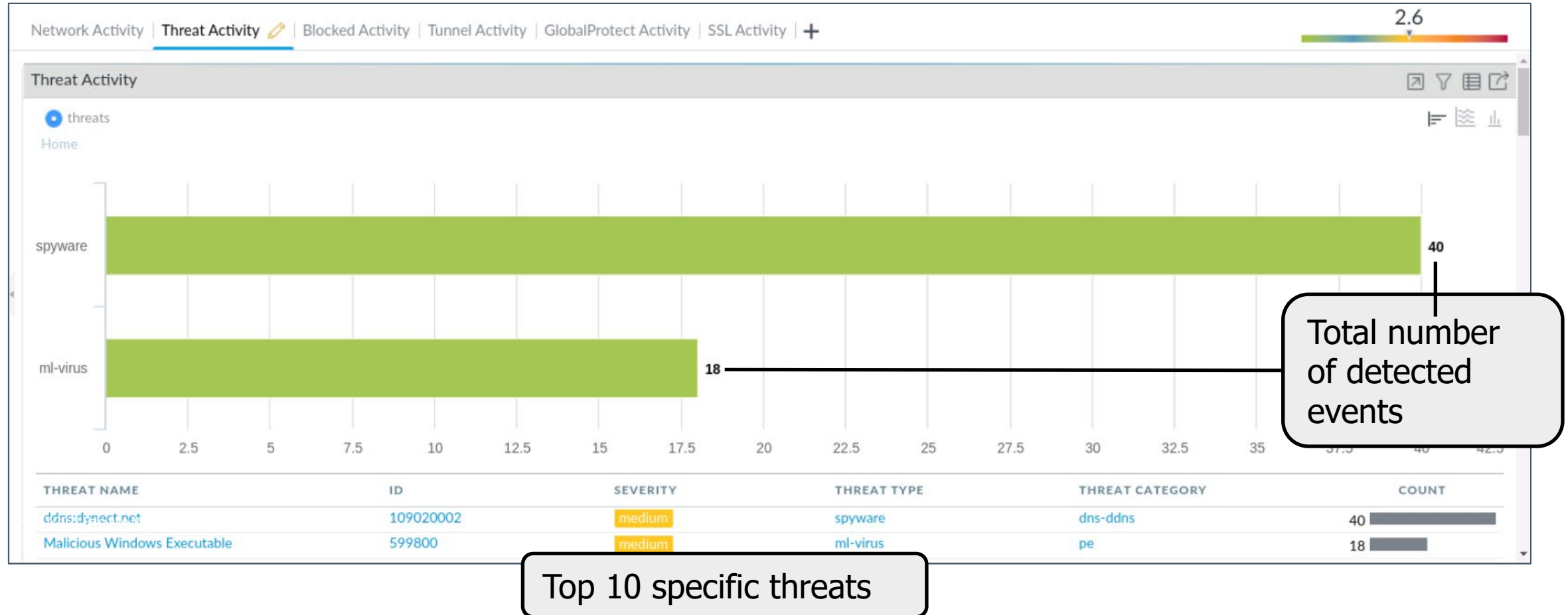
- **Threat Activity**
- **WildFire Activity By File Type**
- **WildFire Activity By Application**
- **Host Visiting Malicious URLs**
- **Host Resolving Malicious Domains**
- **Applications Visiting Non-Standard Ports**
- **Rules Allowing Apps On Non-Standard Ports**
- **Compromised Hosts**

The **Blocked Activity** widgets are helpful to show what has been prevented by the firewall.

For a Tech Doc about this topic, log into Live and search for "ACC Tabs"

Example: Threat Activity Widget

ACC



View threat and traffic information:

In the Dashboard

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In App Scope reports

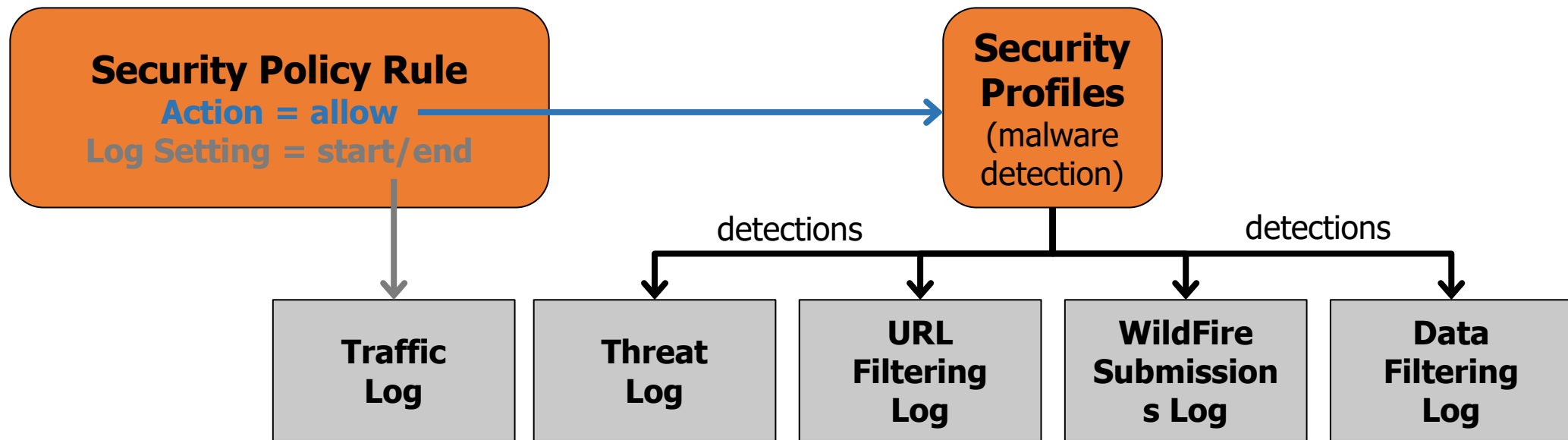
In predefined reports

In custom reports

Forward threat and traffic information to external services



Firewall Logging Overview



- A log contains timestamped information that provides a record of events:
 - Each entry contains a list of artifacts arranged in columns.
- Security policy rules determine what is logged to the Traffic log:
 - Log at session start or session end (end is recommended).
- Security policy and Security Profiles determine what is logged to the other logs.

Example: Traffic Log

Monitor > Logs > Traffic

		RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	DESTINATION	TO PORT	APPLICATION	ACTION	RULE	SESSION END REASON	BYTES
		07/27 17:34:13	deny	Users_Net	Internet	192.168.1.254	34.96.84.34	443	ssl	allow	Users_to_Internet	decrypt-error	6.7k
		07/27 17:34:07	deny	Users_Net	Internet	192.168.1.20	172.217.6.174	80	google-base	reset-both	interzone-default	policy-deny	433
		07/27 17:34:00	deny	Users_Net	Internet	192.168.1.20	172.217.6.174	80	google-base	reset-both	interzone-default	policy-deny	433
		07/27 17:33:58	deny	Users_Net	Internet	192.168.1.20	172.217.6.174	80	google-base	reset-both	interzone-default	policy-deny	433
		07/27 17:33:57	end	Users_Net	Extranet	192.168.1.254	192.168.50.80	80	web-browsing	allow	Users_to_Extranet	tcp-fin	999
		07/27 17:33:57	end	Users_Net	Extranet	192.168.1.254	192.168.50.80	80	web-browsing	allow	Users_to_Extranet	tcp-fin	1.0k
		07/27 17:33:51	deny	Users_Net	Internet	192.168.1.20	172.217.6.174	80	google-base	reset-both	interzone-default	policy-deny	433
		07/27 17:33:49	deny	Users_Net	Internet	192.168.1.20	172.217.6.174	80	google-base	reset-both	interzone-default	policy-deny	433

- Displays an entry for each firewall session:
 - No entry for *in-progress* sessions if **Log Setting = Log at Session End**
- Useful to determine applications seen by the firewall and to improve Security policy

Example: Threat Log

Monitor > Logs > Threat

		RECEIVE TIME	TYPE	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLICATION	ACTION	SEVERITY	FILE NAME
		07/23 16:43:24	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:43:05	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:42:57	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:42:48	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:42:42	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:42:35	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe
		07/23 16:42:04	ml-virus	Malicious Windows Executable	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	reset-both	medium	wildfire-test-pe-file.exe

- Displays entries when threats are detected in allowed traffic.
- *Critical*-severity and *high*-severity threats always should be blocked.
- *Medium*-severity threats might need to be blocked.
- *Low*-severity and *informational*-severity threats should generate an alert.

Example: URL Filtering Log

Monitor > Logs > URL Filtering

	RECEIVE TIME	CATEGORY	URL CATEGORY LIST	URL	FROM ZONE	TO ZONE	SOURCE	DESTINATION	APPLICATION	ACTION
	07/27 17:22:02	block-per-company-policy	block-per-company-policy,news,low-risk	www.theguardian.com/favicon.ico	Users_Net	Internet	192.168.1.20	151.101.129.111	web-browsing	block-url
	07/27 17:22:01	block-per-company-policy	block-per-company-policy,news,low-risk	www.theguardian.com/login/css/latofonts....	Users_Net	Internet	192.168.1.20	151.101.129.111	web-browsing	block-url
	07/27 17:22:01	block-per-company-policy	block-per-company-policy,news,low-risk	www.theguardian.com/	Users_Net	Internet	192.168.1.20	151.101.129.111	web-browsing	block-url
	07/27 17:21:11	block-per-company-policy	block-per-company-policy,news,low-risk	www.nbcnews.com/favicon.ico	Users_Net	Internet	192.168.1.20	23.64.169.49	web-browsing	block-url
	07/27 17:21:11	block-per-company-policy	block-per-company-policy,news,low-risk	www.nbcnews.com/login/css/latofonts.css	Users_Net	Internet	192.168.1.20	23.64.169.49	web-browsing	block-url
	07/27 17:21:11	block-per-company-policy	block-per-company-policy,news,low-risk	www.nbcnews.com/	Users_Net	Internet	192.168.1.20	23.64.169.49	web-browsing	block-url
	07/27 17:21:05	block-per-company-policy	block-per-company-policy,news,low-risk	www.nbcnews.com/favicon.ico	Users_Net	Internet	192.168.1.20	23.64.169.49	web-browsing	block-url

- Displays entries for URLs or URL categories configured in URL Filtering Profiles:
 - Configure the website or URL category with at least the “alert” action.
- Use the information to improve your Security policy and URL Filtering Profiles.

Example: WildFire Submissions Log

Monitor > Logs > WildFire Submissions

	RECEIVE TIME	FILE NAME	SOURCE ZONE	DESTINA... ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	DEST... PORT	APPLICATION	RULE	VERDICT	ACTION	SEVERITY	FILE TYPE
	07/23 21:08:22	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 20:44:54	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 20:10:31	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 20:10:31	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 19:52:18	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 19:36:17	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 18:49:32	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe
	07/23 18:41:32	wildfire-test-pe-file.exe	Users_Net	Internet	192.168.1.20	35.222.124.72	80	web-browsing	Users_to_Internet	malicious	allow	high	pe

- Displays submissions by the firewall to WildFire
- Use to determine:
 - Who sent the malware (can it be blocked?)
 - Who received the malware (who potentially is infected?)
 - The filename or URL link used to deliver the malware (what to remove or block)

Example: Data Filtering Log

Monitor > Logs > Data Filtering

		RECEIVE TIME	CATEGORY	FILE NAME	THREAT ID/NAME	FROM ZONE	TO ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	TO PORT	APPLICATION	ACTION
		07/17 17:55:24	private-ip-addresses	webmail.php	Hypertext Preprocessor PHP File	Users_Net	Extranet	192.168.1.20	192.168.50.150	80	squirrelmail	alert
		07/17 17:55:16	private-ip-addresses	login.php	Hypertext Preprocessor PHP File	Users_Net	Extranet	192.168.1.20	192.168.50.150	80	web-browsing	alert
		07/17 17:51:58	computer-and-internet-info	04457f5911080bb0...	Unknown Binary File	Extranet	Internet	192.168.50.150	91.189.91.38	80	apt-get	alert
		07/17 17:51:44	computer-and-internet-info	Packages.gz	GZIP	Extranet	Internet	192.168.50.150	104.207.151.13	80	apt-get	alert
		07/17 17:51:34	computer-and-internet-info	ec430cd4d2899367...	Unknown Binary File	Extranet	Internet	192.168.50.150	91.189.91.39	80	apt-get	alert

- Displays entries when sensitive files or data are seen by the firewall:
 - Configure the File Blocking or Data Filtering Profile with at least the “alert” action.
- Use the information:
 - To improve the security configuration
 - In incident response

Example: Unified Log

Monitor > Logs > Unified

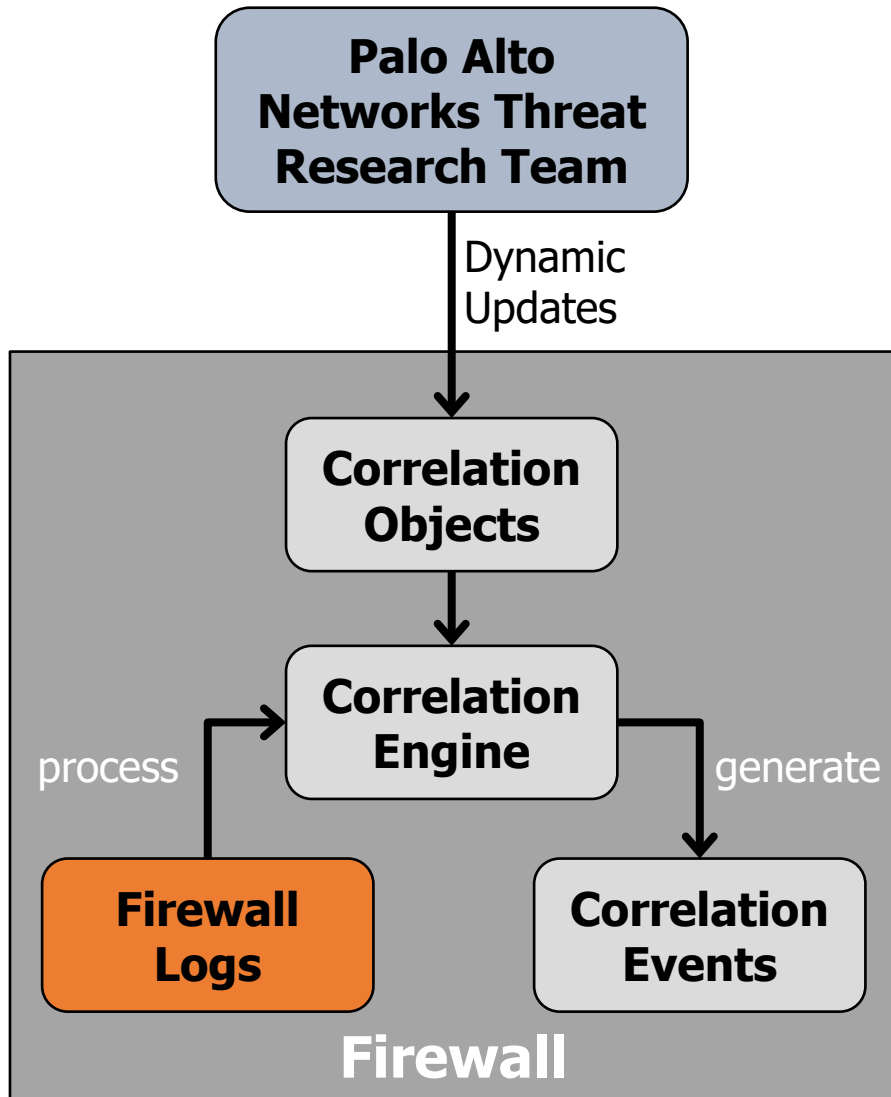
Show Effective Queries

	LOG TYPE	FILTER
☒	traffic	N/A
☒	threat	N/A
☒	url	N/A
☒	data	N/A
☒	wildfire	N/A
☒	tunnel	N/A
☒	auth	N/A
☒	iptag	N/A
☒	globalprotect	N/A
☒	decryption	N/A

	LOG TYPE	RECEIVE TIME	LOG SUBTYPE	SESSION ID	SOURCE ZONE	DESTINATION ZONE	SOURCE ADDRESS	DESTINATION ADDRESS	DESTINATION PORT	APPLICATION	ACTION	RULE	BYTES
	traffic	07/27 17:53:39	deny	68840	Users_Net	Internet	192.168.1.20	172.217.6.142	80	google-base	reset-both	interzone-default	433
	traffic	07/27 17:53:37	deny	68837	Users_Net	Internet	192.168.1.20	172.217.12.78	80	google-base	reset-both	interzone-default	433
	traffic	07/27 17:53:37	end	68808	Users_Net	Internet	192.168.1.20	4.2.2.2	53	dns	allow	Users_to_Internet	79
	traffic	07/27 17:53:35	end	68796	Users_Net	Extranet	192.168.1.254	192.168.50.53	53	dns	allow	Users_to_Extranet	712
	traffic	07/27 17:53:35	end	68796	Users_Net	Extranet	192.168.1.254	4.2.2.2	53	dns	allow	Users_to_Internet	196
	traffic	07/27 17:53:35	end	68796	Users_Net	Extranet	192.168.1.25	192.168.50.53	53	dns	allow	Users_to_Extranet	592
	traffic	07/27 17:53:35	end	68796	Users_Net	Extranet	192.168.1.254	4.2.2.2	53	dns	allow	Users_to_Internet	556

- View several log types from a single location:
 - Can modify the list of included log types
- Simplifies threat and traffic investigation and analysis

Correlation Engine, Objects, and Events



- Automated utility to detect and report possibly infected hosts
- The automated *correlation engine*:
 - Is an analytics tool
 - Examines firewall logs
 - Looks for behavior patterns that match criteria defined in *correlation objects*:
 - Objects define:
 - Pattern to match
 - Logs to reference
 - Time period to look at
- Generates *correlation events*:
 - Notices of possible infection

View threat and traffic information:

In the Dashboard

In the ACC

In the logs

In App Scope reports

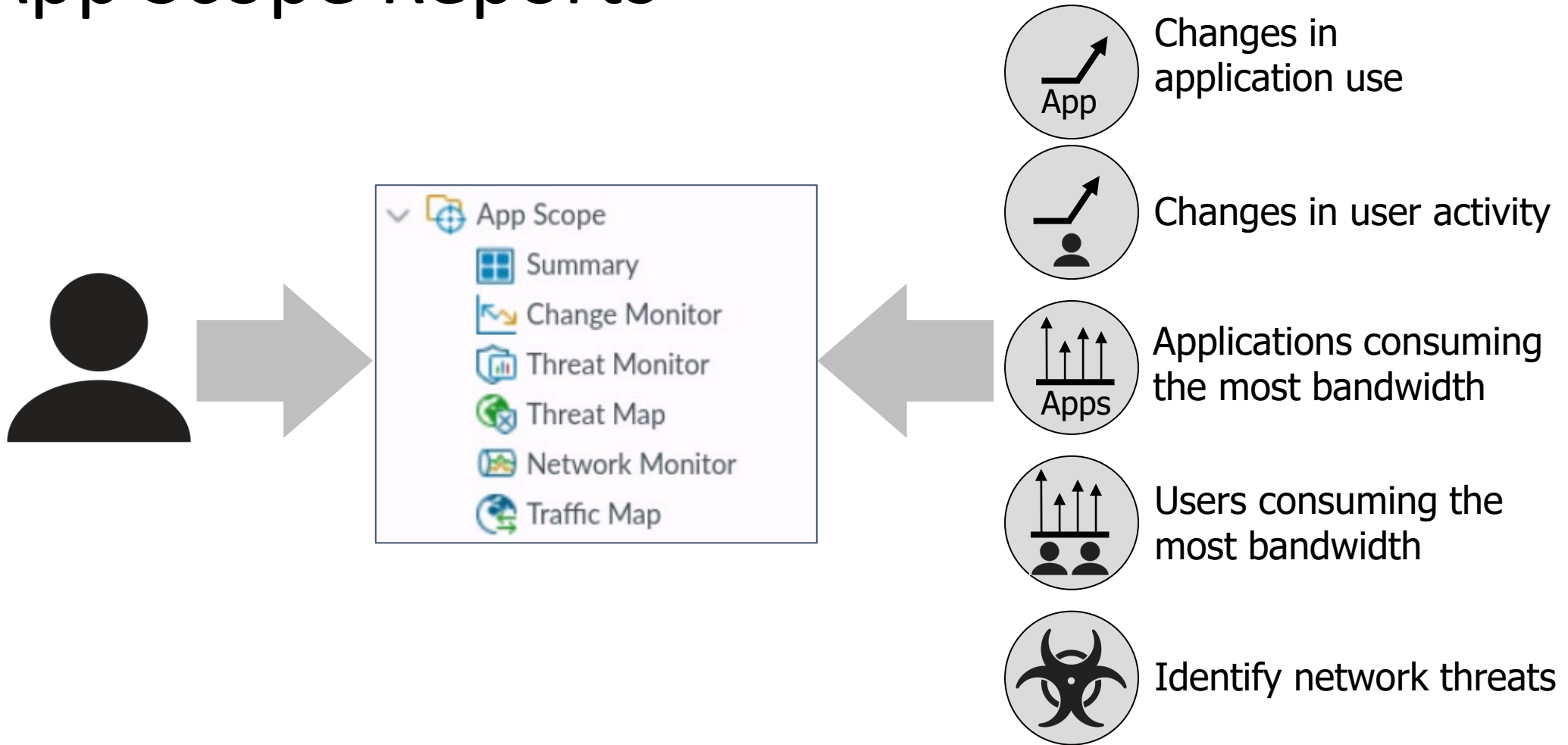
In predefined reports

In custom reports

Forward threat and traffic information to external services



App Scope Reports



App Scope Reports: What's Available?

Report Name	Description
Summary Report	Set of six graphical reports displayed in a single browser pane
Top 5 Gainers	Applications showing largest increase in number of sessions (last 60 minutes)
Top 5 Losers	Applications showing largest decrease in number of sessions (last 60 minutes)
Top 5 Bandwidth Consuming Source	Devices sending the most bytes of data (last 60 minutes.)
Top 5 Bandwidth Consuming Apps	Applications sending the most bytes of data (last 24 hours)
Top 5 Bandwidth Consuming App categories	Application categories sending the most bytes of data (last 24 hours)
Top 5 Threats	Threats encountered the most often (last 24 hours)

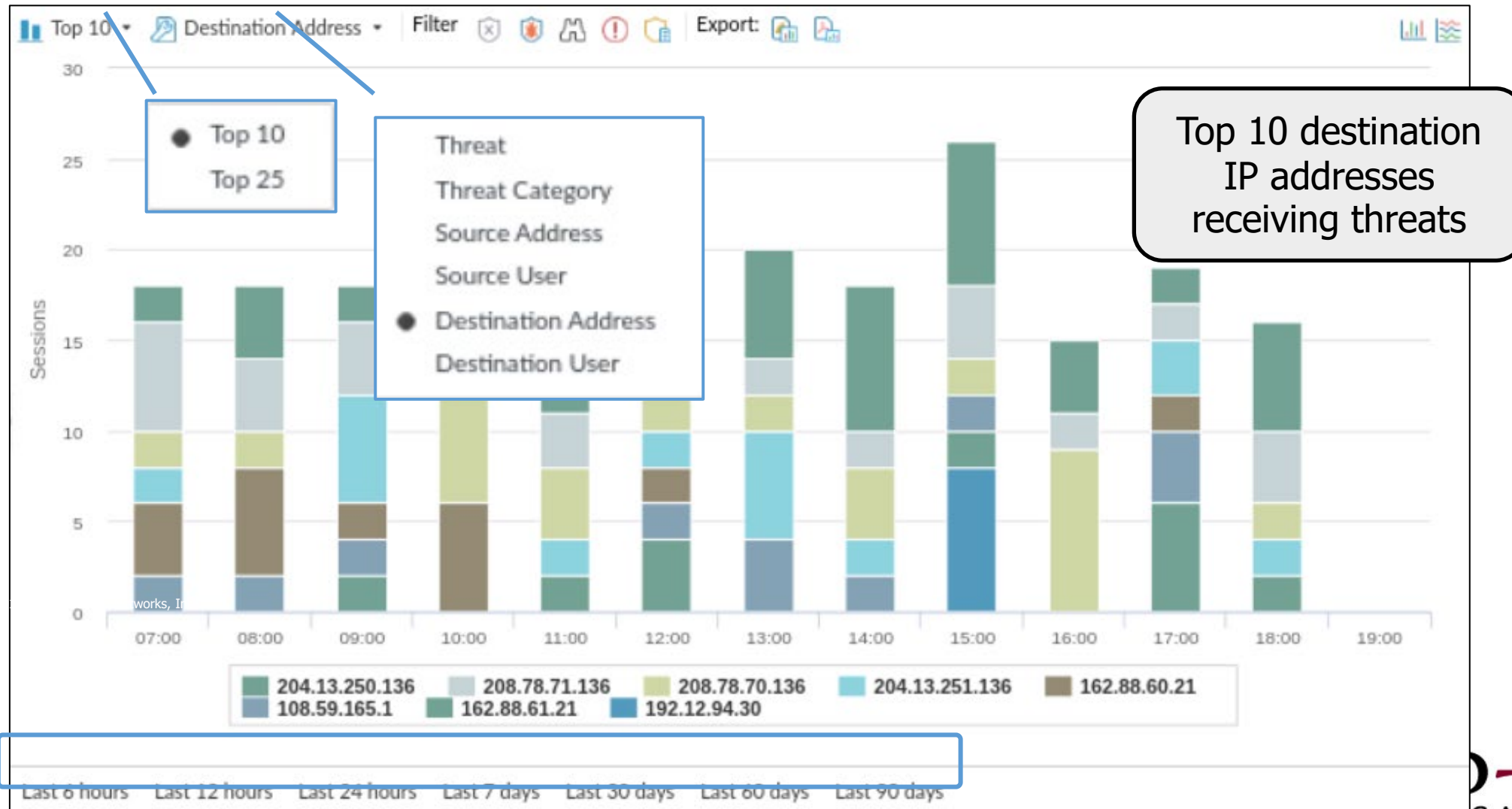
App Scope Reports: What's Available? (Cont.)

Report Name	Description
Change Monitor	Applications, users, sources, and destinations showing the largest increase in the number of sessions over the selected time period
Threat Monitor	Threats appearing in the largest number of sessions over the selected time period (by threat name, category, source, or destination)
Threat Map	World map showing sources and destinations of threats (by country)
Network Monitor	Applications, users, sources, and destinations showing the largest network bandwidth consumption over the selected time period
Traffic Map	World map showing sources and destinations of traffic (by country)

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Example: App Scope Report

Monitor > App Scope > Threat Monitor



Top 10 destination IP addresses receiving threats

Last 6 hours Last 12 hours Last 24 hours Last 7 days Last 30 days Last 60 days Last 90 days

View threat and traffic information:

In the Dashboard

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In App Scope reports

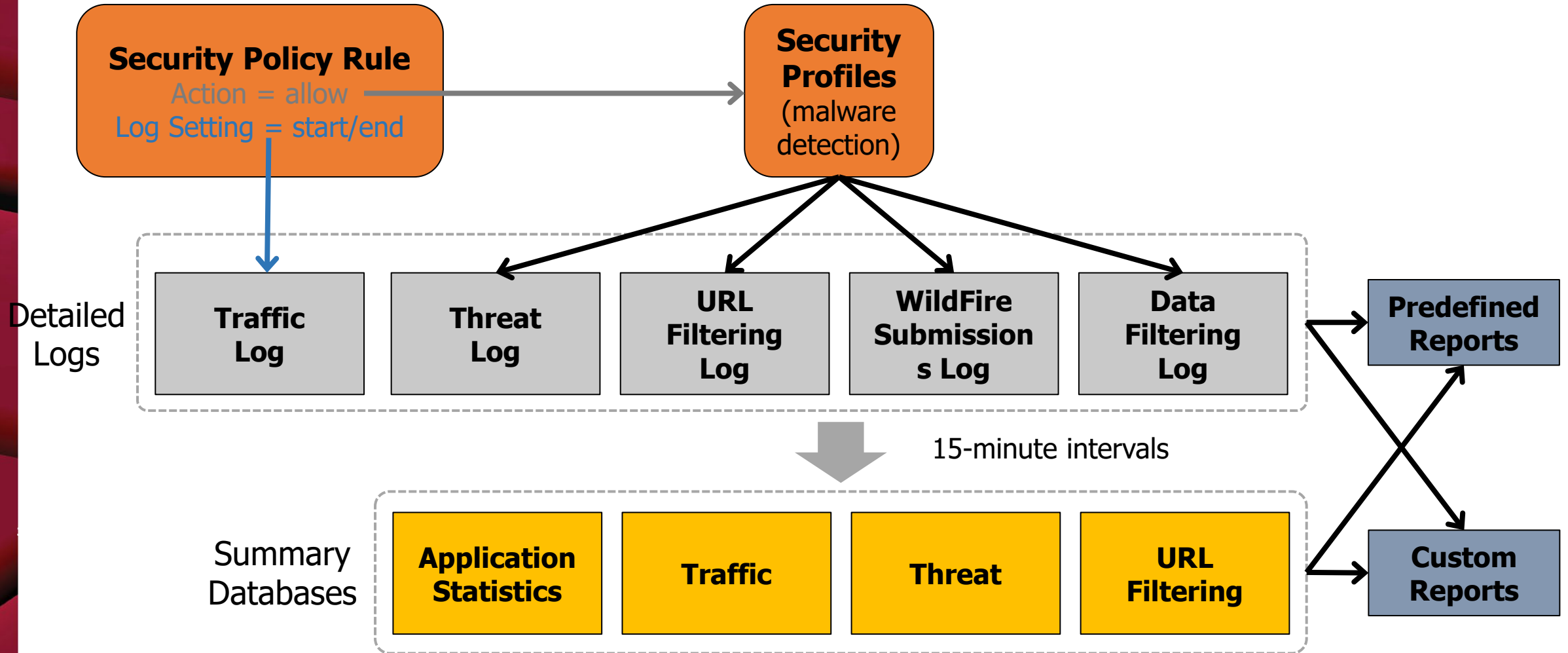
In predefined reports

In custom reports



Forward threat and traffic information to external services

Firewall Logging and Reporting Overview



Predefined Reports

Monitor > Reports

Click report to view in web interface.

Click to download as a PDF.

Application Reports —

- New Applications
- Applications
- Application Categories
- Technology Categories
- HTTP Applications
- SaaS Application Usage
- Denied Applications

Traffic Reports +

- Security Rules
- Sources
- Source Countries
- Destinations
- Destination Countries
- Connections
- Source Zones
- Destination Zones
- Ingress Interfaces
- Egress Interfaces
- Denied Sources
- Denied Destinations

Threat Reports —

- Botnet
- Threats
- Inline Cloud Analysis
- Attacker Sources
- Attacker Destinations
- Attackers By Source Country
- Attackers By Destination Country
- Victim Sources
- Victim Destinations
- Victims By Source Countries
- Victims By Destination Countries

URL Filtering Reports —

- URL Categories
- Inline Categorization Verdicts
- URL Users
- URL User Behavior
- Web Sites
- Blocked Categories
- Blocked Users
- Blocked User Behavior
- Blocked Sites
- Credential Post Detected

PDF Summary Reports —

- predefined

January 2022

S	M	T	W	T	F	S
26	27	28	29	30		
2	3	4	5	6		
9	10	11	12	13		
16	17	18	19	20		
23	24	25	26	27		
30	31	1	2	3		

- More than 40 predefined reports.
- Arranged in five categories.
- Firewall generates a new report each day.
- To display a report:
 - Click a date.
 - Click a report.

Example: Threats Report

Monitor > Reports

	THREAT ID/NAME	ID	THREAT/CONTENT TYPE	COUNT		Application Reports +
1	Suspicious TLS Evasion Found	14978	spyware	380		Traffic Reports +
2	Suspicious Domain	12000000	spyware	236		Threat Reports -
3	Suspicious HTTP Evasion Found	14984	spyware	174		Botnet
4	41101	41101	vulnerability	1		Threats
5	41100	41100	vulnerability	1		Attacker Sources
						Attacker Destinations
						Attackers By Source Countries
						Attacker Countries
						Victim Destinations
						Victim Destinations
						URL Filtering Reports +
						PDF Summary Reports +
						< June 2020 >
						S M T W T F S
						31 1 2 3 4 5 6
						7 8 9 10 11 12 13
						14 15 16 17 18 19 20
						21 22 23 24 25 26 27
						28 29 30 1 2 3 4
						5 6 7 8 9 10 11
						Export to PDF Export to CSV Export to XML

Select.



View threat and traffic information:

In the Dashboard

In the ACC

In the logs

In App Scope reports

In predefined reports

In custom reports



Forward threat and traffic information to external services

Custom Reports

- Custom reports show only the information you want:
 - You choose the logs and log columns to view.
- Schedule reports or run-on demand.
- View a report:
 - In the web interface
 - By exporting a PDF, CSV, or XML file

Monitor > Manage Custom Reports

The screenshot shows the 'Manage Custom Reports' interface. At the top, there is a search bar and a '1 item' indicator with a right arrow and a close 'X' button. Below this is a table with the following columns: NAME, DESCRIPTION, DATABASE, TIME FRAME, ROWS, SORT BY, GROUP BY, and SCHEDULED. The first row in the table is for a report named 'Test Report', which has a description of 'Test Report for Application Statistics', is from the 'Application Statistics' database, covers the 'Last 30 Days' time frame, has 10 rows, is sorted by 'Sessions', and is grouped by 'category-of-name'. The 'SCHEDULED' column for this report has an unchecked checkbox. Below the table, there is a row of action buttons: '+ Add', '- Delete', and 'Clone'. Three callout boxes are present: one pointing to the 'Test Report' name with the text 'Display a custom report.', one pointing to the '+ Add', '- Delete', and 'Clone' buttons with the text 'Create, delete, or clone a report.', and one pointing to the 'Test Report' row with the text 'Display a custom report.'

☐	NAME	DESCRIPTION	DATABASE	TIME FRAME	ROWS	SORT BY	GROUP BY	SCHEDULED
☐	Test Report	Test Report for Application Statistics	Application Statistics	Last 30 Days	10	Sessions	category-of-name	☐

+ Add - Delete Clone

Custom Report Example

Monitor > Manage Custom Reports

Custom Report

Report Setting | Test Report (100%) x

	APP CATEGORY	APP CONTAINER	APPLICATION NAME	DAY RECEIVED	RISK	SESSIONS	THREATS
1	networking		dns	Sun, Jul 26, 2020	3	72.1k	0
2	networking		dns	Sat, Jul 25, 2020	3	68.2k	0
3	networking		dns	Fri, Jul 24, 2020	3	65.8k	0
4	networking		dns	Mon, Jul 27, 2020	3	53.9k	0
5	networking		dns	Thu, Jul 23, 2020	3	18.0k	0
6	networking		dns	Fri, Jul 17, 2020	3	2.3k	0
7	unknown		insufficient-data	Fri, Jul 24, 2020	1	1.4k	0
8	unknown		insufficient-data	Sat, Jul 25, 2020	1	1.4k	0
9	unknown		insufficient-data	Sun, Jul 26, 2020	1	1.4k	0
10	unknown		insufficient-data	Mon, Jul 27, 2020	1	1.2k	0

Export to PDF Export to CSV Export to XML

Sort By Sessions Top 10

Group By App Category 10 Groups

Time Frame Last 30 Days

Database Application Statistics

Selected Columns

- App Category
- App Container
- Application Name
- Day
- Risk

Custom Report with a Query Builder Filter

Monitor > Manage Custom Reports > Add

The screenshot shows the 'Custom Report' configuration window. The 'Report Setting' tab is active, displaying fields for Name, Description, Database, Scheduled status, Time Frame, Sort By, and Group By. The 'Available Columns' list includes App Category, App Sub Category, App Technology, Bytes, and Device Name. The 'Selected Columns' list includes App Container, Application Name, Day, Risk, and Sessions. The 'Query Builder' section at the bottom contains the filter '(category-of-name eq networking)'. A callout box points to this filter with the text: 'Include only log entries where **networking** is the category of name.'

Custom Report

Report Setting

Load Template → Run Now

Name: Test Report

Description: Test Report for Application Statistics

Database: Application Statistics

☒ Scheduled

Time Frame: Last 30 Days

Sort By: Sessions Top 10

Group By: App Category 10 Groups

Available Columns

- App Category
- App Sub Category
- App Technology
- Bytes
- Device Name

Selected Columns

- App Container
- Application Name
- Day
- Risk
- Sessions

Query Builder

(category-of-name eq networking)

Filter Builder

Include only log entries where **networking** is the category of name.

- **Query Builder** applies filters to source log.
- Only log entries matching the filter are included in the report.
- Filters match only if a specific *value* is in a column.
- This example reports only applications where the category of name equals networking.

Query Builder Report Example

Monitor > Manage Custom Reports

Custom Report ?

Report Setting | Test Report (100%) ×

	APP CATEGORY	APPLICATION NAME	DAY RECEIVED	RISK	SESSIONS	THREATS
1	networking	dns	Sun, Jul 26, 2020	3	72.1k	0
2		dns	Sat, Jul 25, 2020	3	68.2k	0
3		dns	Fri, Jul 24, 2020			0
4		dns	Mon, Jul 27, 2020			0
5		dns	Thu, Jul 23, 2020			0
6		dns	Fri, Jul 17, 2020			0
7		ssl	Thu, Jul 23, 2020	4	66	0
8		netbios-dg	Fri, Jul 17, 2020	2	66	0
9		ssl	Fri, Jul 17, 2020	4	33	0
10		ntp	Fri, Jul 17, 2020	2	32	0

Applications with the App Category of networking

Export to PDF Export to CSV Export to XML

- Report provides intelligence to configure your perimeter Security policy rules.
- Modify the app category name in the filter to see applications from other application categories.

View threat and traffic information:

In the Dashboard

In the ACC

In the logs

In App Scope reports

In predefined reports

In custom reports

Forward threat and traffic information to external services



Device Telemetry

- Shared data with Palo Alto Networks.
- Data is used to improve visibility into device health, performance, capacity planning, and configuration.
- Requires a Cortex Data Lake license.

Device > Setup > Telemetry

Management	Operations	Services	Interfaces	Telemetry	Content-ID	WildFire	Session	HSM
------------	------------	----------	------------	-----------	------------	----------	---------	-----

Telemetry

Threat Prevention

Device Health and Performance

Product Usage

Status success

Reason

device-health-performance

Status success

Reason

Last Attempt Thu Jan 23 15:19:04 PST 2020

Last Success Thu Jan 23 15:19:04 PST 2020

No. of Failed Attempts 0

threat-prevention

Status success

Configure Device Telemetry

Device > Setup > Telemetry

The screenshot shows the 'Telemetry' configuration page. It has a title bar 'Telemetry' with a help icon. Below is a 'Telemetry Sharing' section with explanatory text. The 'Settings' section is active, showing checkboxes for 'Enable Telemetry' and three categories: 'Threat Prevention', 'Device Health and Performance', and 'Product Usage'. A 'Telemetry Region' dropdown is set to 'Americas'. At the bottom is a 'Generate Telemetry File' button. Four callout boxes with arrows point to specific elements: 'Select which categories of data to share.' points to the category checkboxes; 'Select Region to enable sharing of device telemetry data.' points to the 'Americas' dropdown; 'Download a live example of data collected and shared.' points to the 'Generate Telemetry File' button; and an unlabeled callout points to the 'Telemetry Sharing' text area.

Telemetry

Telemetry Sharing

The analysis of telemetry data provides information that increases visibility into the usage and performance of Palo Alto Networks products and services. Palo Alto Networks uses the data from your systems to improve threat prevention research, to analyze device utilization and performance, and to offer you insights intended to maximize the value you obtain from Palo Alto Networks products.

You must select a region to enable telemetry collection. Once selected, you can enable or disable telemetry collection at any time using the settings below. The information you share might include personal information. You can view the details of what is collected by clicking on the icon beside each telemetry category. You can also see the actual data that will be sent to Palo Alto Networks by clicking Generate Telemetry File at the bottom of this screen. [Learn more](#) about Palo Alto Networks telemetry and see telemetry privacy policies in the [Privacy Data Sheet](#).

All telemetry data is sent to Cortex Data Lake. If your organization currently has a Cortex Data Lake license, your telemetry region selection choice is restricted to your Cortex Data Lake region.

Settings

☒ **Enable Telemetry**

☒ **Threat Prevention**
Includes URL Filtering and Threat Prevention summaries

☒ **Device Health and Performance**
Includes resource utilization (CPU/Memory/Sessions etc.)

☒ **Product Usage**
Includes configuration

Telemetry Region: **Americas**
Select Region to enable telemetry

Generate Telemetry File

Select which categories of data to share.

Select Region to enable sharing of device telemetry data.

Download a live example of data collected and shared.

Monitor Device Telemetry

Device > Setup > Telemetry

The screenshot shows the Telemetry configuration page with the following widgets:

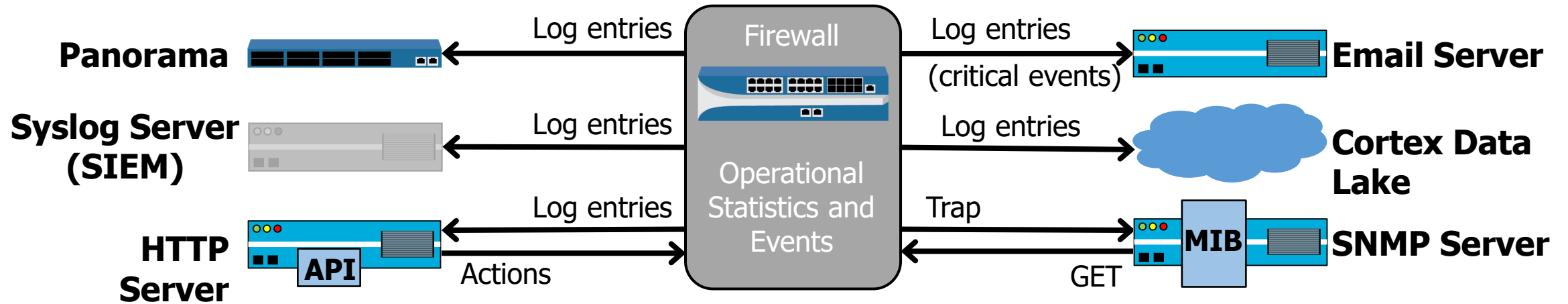
- Threat Prevention**: Status: success, Reason: (empty), Last Attempt: Thu Jan 23 15:19:04 PST 2020, Last Success: Thu Jan 23 15:19:04 PST 2020, No. of Failed Attempts: 0.
- Device Health and Performance**: Status: success, Reason: (empty), Last Attempt: Thu Jan 23 15:19:04 PST 2020, Last Success: Thu Jan 23 15:19:04 PST 2020, No. of Failed Attempts: 0.
- Product Usage**: Status: success, Reason: (empty), Last Attempt: Thu Jan 23 15:19:04 PST 2020, Last Success: Thu Jan 23 15:19:04 PST 2020, No. of Failed Attempts: 0.

Callout 1: Widgets display Status for each category selected.

Callout 2: If transmission fails, firewall waits 10 minutes before attempting to resend the data.

Firewall Log Forwarding Review

Supported External Log Destinations



- Benefits of log forwarding and centralized logging:
 - Better log availability (off-firewall backup)
 - Centralized log analysis (PAN-OS software or third-party applications)
 - Longer log retention
 - Alerts for critical events
 - Automated responses (via Web API)

Configure a Server Profile: Syslog Example

- Create a Server Profile:
 - Defines where and how the firewall should connect to an external service

Device > Server Profiles > Syslog > Add

Syslog Server Profile

Name

Syslog-Log-Forwarding

Servers

Custom Log Format

NAME	SYSLOG SERVER	TRANSPORT	PORT	FORMAT	FACILITY
Syslog1	192.168.50.10	UDP	514	BSD	LOG_USER

+ Add

- Delete

Enter the IP address or FQDN of the Syslog server

How to format forwarded log entries

LOG_USER

LOG_LOCAL0

LOG_LOCAL1

LOG_LOCAL2

LOG_LOCAL3

LOG_LOCAL4

LOG_LOCAL5

LOG_LOCAL6

LOG_LOCAL7

Configure Logs to Forward: Example

Defines which logs or log entries to forward to which external services

Objects > Log Forwarding > Add

Log Forwarding Profile

Name

Forwarding-Threat-Information

Description

Forwarding Threat Log Information to the Syslog Server

4 items

☐	NAME	LOG TYPE	FILTER	FORWARD METHOD	BUILT-IN ACTIONS
☐	Forwarding-Traffic-Logs	threat	All Logs	<u>SysLog</u> Syslog-Log-Forwarding	
☐	Forwarding-Threat-Logs	threat	(severity eq high)	<u>SysLog</u> Syslog-Log-Forwarding	
☐	Forwarding-WildFire-Logs	wildfire	All Logs	<u>SysLog</u> Syslog-Log-Forwarding	
☐	Forwarding-URL-Logs	url	All Logs	<u>SysLog</u> Syslog-Log-Forwarding	

Server Profile

Means all log *entries* (logs are *not* filtered)

Apply Log Forwarding

Policies > Security > <rule>

The screenshot shows the 'Log Setting' section of a security rule configuration. It includes a 'Log at Session Start' checkbox (unchecked), a 'Log at Session End' checkbox (checked), and a 'Log Forwarding' dropdown menu set to 'Forwarding-Threat-Information'. The 'Forwarding-Threat-Information' option is highlighted with a black box.

- Log forwarding applied per Security policy rule
- Forwards logs for traffic matching:
 - Security policy rule
 - Log Forwarding Profile settings

Network > Zones > <zone>

The screenshot shows the 'Zone' configuration page. It includes a 'Name' field set to 'Users_Net', a 'Log Setting' dropdown menu set to 'Forwarding-Threat-Information', and a 'Type' dropdown menu set to 'Layer3'. Below these fields is a section titled 'INTERFACES' with a list of interfaces, including 'ethernet1/2'. The 'Log Setting' dropdown is highlighted with a black box.

- Log forwarding applied per security zone
- Forwards threats detected by the Zone Protection Profile

Module Summary

Now that you have completed this module, you should be able to:

- Monitor threat and traffic information using the Dashboard and the ACC
- Monitor threat and traffic information using the logs
- Monitor threat and traffic information using App Scope reports
- Monitor threat and traffic information using the botnet report
- Monitor threat and traffic information using predefined and custom reports
- Configure firewall log forwarding to external services

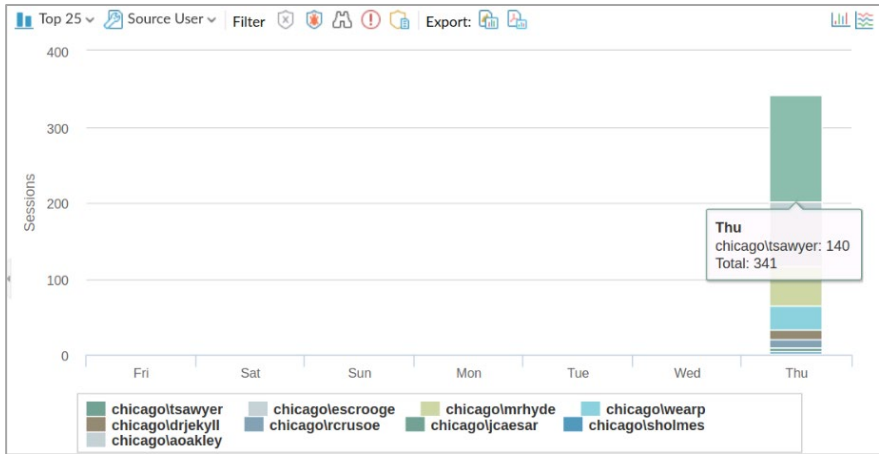


Questions



Lab 13: Overview

User ID Reports



Application Reports

	APPLICATION NAME	BYTES	SESSIONS
1	dns	22.8M	71.9k
2	web-browsing	2.5M	1.7k
3	netbios-dg	656.3k	1.2k
4	google-base	1.6M	892
5	paloalto-updates	114.9M	643
6	paloalto-wildfire-cloud	6.6M	633
7	ntp	44.8k	174
8	paloalto-dns-security	6.6M	35
9	ssl	1.2M	22
10	apt-get	12.0M	9
11	pan-db-cloud	13.1M	5
12	ping	196	1
13	netbios-ns	37.9k	1
14	ssh	36.2k	1

Predefined Reports

Application Reports	+
Traffic Reports	-
Security Rules	
Sources	
Source Countries	
Destinations	
Destination Countries	
Connections	
Source Zones	
Destination Zones	
Ingress Interfaces	
Egress Interfaces	
Denied Sources	
Denied Destinations	
Unknown TCP Sessions	
Unknown UDP Sessions	
Threat Reports	+
URL Filtering Reports	+
PDF Summary Reports	+

URL Filtering Reports

	URL DOMAIN	CATEGORY	COUNT
1	www.amazon.com	shopping	29
2	www.hackthissite.org	hacking	28
3	www.taobao.com	shopping	20
4	www.tmall.com	shopping	20
5	global.jd.com	shopping	20
6	www.aliexpress.com	shopping	20
7	shodan.io	hacking	11
8	www.shutterfly.com	shopping	10

Lab 13: Locating Valuable Information Using Logs and Reports

- View threat information using the Dashboard
- View application information using the Dashboard
- View threat information using the ACC
- View application information using the ACC
- View threat information using the Threat log
- View application information using the Traffic log
- View threat information using App Scope reports
- View threat information using predefined reports
- View application information using predefined reports
- View threat and application information using custom reports



**Protecting our
digital way
of life.**



In Class Lab

- Inspect Traffic Log CSV
 - Log.csv on blackboard
- Anything Interesting?
- Did you see anything that looks bad?
- Can you find any email rows?
- Can you find any rows that seem to be benign web surfing?

End Content

Unit 13 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 14 Lab: Palo Alto Lab 13: Using Decryption to Block Threats in Encrypted Traffic
- Unit 13 Assignment 1: Github Profile
- Unit 13 Assignment 2: LinkedIn Profile
- Unit 13 Quiz